

REVIEW

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Artificial intelligence in automated semen analysis: a systematic review of technical performance and clinical validity

Ravi Yadav^{1*} and Debasrita Banerjee¹

Abstract

Background Male infertility accounts for approximately 40–50% of all infertility cases worldwide, with semen analysis remaining the cornerstone of diagnosis. Manual semen evaluation is limited by substantial inter- and intra-observer variability, particularly for sperm morphology assessment, which has motivated the development of artificial intelligence (AI)-based automated systems.

Objective To systematically evaluate the *technical performance* of AI-based automated semen analysis systems, specifically for sperm morphology classification and motility assessment compared to manual expert evaluation, and to appraise the quality of evidence and identify critical gaps in clinical validation. DNA fragmentation prediction and zona pellucida (ZP) binding capability prediction is additionally examined as emerging, non-standard applications.

Methods A systematic search of PubMed, Scopus, and Web of Science (January 2015 – November 2025) was conducted following PRISMA 2020 guidelines. Studies comparing deep learning or machine learning methods to manual expert assessment or conventional CASA systems for human semen analysis were included. Risk of bias was assessed using QUADAS-2, RoB 2, and ROBINS-I; evidence certainty was graded using GRADE methodology. It should be noted that manual WHO semen analysis, used as the reference standard across most included studies, carries well-documented reproducibility limitations, particularly for morphology, which constrains the interpretability of AI performance metrics.

Results Eighteen eligible studies (15 observational, 3 validation) were identified. Deep learning models, particularly ResNet-50 variants with attention mechanisms (CBAM) and ensemble strategies, achieved morphology classification accuracy of 55–96.77% (mean: 82.4%). CNN-based motility assessment demonstrated strong correlation with manual WHO classification (Pearson $r=0.88$ – 0.89). Emerging AI applications for DNA fragmentation prediction (MAE: 0.05–0.10) and ZP binding capability (accuracy: 96.7%) showed technical feasibility but limited clinical outcome validation. One multi-centre study demonstrated model performance consistency across laboratory sites; however, robust external validation across diverse settings remains limited. GRADE certainty of evidence ranged from low to moderate across outcomes.

*Correspondence:

Ravi Yadav
Yravi6945@gmail.com

Full list of author information is available at the end of the article

Conclusion AI-based automated semen analysis demonstrates promising technical performance for sperm morphology and motility assessment; however, the current evidence base does not establish AI-based semen analysis as a validated diagnostic or prognostic tool for male infertility. In the absence of prospective, outcome-driven validation studies, these systems should be considered investigational. High technical accuracy relative to a manual reference standard, itself subject to reproducibility limitations, does not confirm clinical validity or improvement in patient outcomes. Future research must prioritise prospective multi-centre validation, standardised training datasets, and randomised controlled trials evaluating the effect of AI-guided sperm selection on clinically meaningful endpoints, including fertilisation rates, implantation success, and live birth outcomes.

Keywords Artificial intelligence, Deep learning, Semen analysis, Male infertility, Sperm morphology, Automated diagnosis, Clinical validity, Systematic review

Introduction

Background

Male factor is responsible for 40–50% of all cases of human infertility worldwide [1]. Although a comprehensive diagnostic evaluation of male factors causing infertility encompasses endocrine, genetic, and anatomical investigations, semen analysis is considered a basic biochemical and microscopic investigation. Semen analysis is standardized by the World Health Organization (WHO) and has been most recently updated in its 6th edition [2]. Manual evaluation of semen helps in the quantification of oligozoospermia, asthenozoospermia, and teratozoospermia. However, manual evaluation is considered a basic investigation and is always plagued by a high degree of inter- and intra-observer variability. For instance, the inter-observer concordance rate for assessing sperm morphology is between 40 and 60%, mainly due to a high degree of subjectivity involved in microscopic evaluation and staining procedures [3].

While Computer-Assisted Semen Analysis (CASA) systems were developed to minimize the subjectivity of the analysis of kinematic parameters, the application of CASA systems in the analysis of semen morphology has shown poor correlation with expert analysis and has been further limited by cost factors [4].

Problem statement

The shortcomings in the traditional semen analysis process are limiting its reproducibility in clinical practice as well as its efficiency in terms of workflow. Manual semen analysis, which demands at least a minimum of 1,000 h of supervised laboratory training, also shows a high degree of subjectivity in the evaluation process, especially in the context of sperm morphology [5]. While the conventional CASA technique can efficiently handle the motility kinetics in semen analysis, the morphological anomalies in the semen are not adequately addressed in the process, especially in the context of standardizing the subjective evaluation of morphological characteristics in semen analysis, which are considered the most difficult to standardize. Moreover, the manual semen analysis technique, as well as the conventional CASA technique, are not equipped

with the facility to evaluate the functional characteristics of semen in real-time, especially in the context of predicting the DNA integrity as well as the zona pellucida binding capability in semen analysis, which are critical in determining the ART outcome.

Research gap

In fact, the application of artificial intelligence (AI) and machine learning (ML) has shown considerable promise in the automation of complex image-based procedures in the laboratory. Nevertheless, the rigorous systematic investigation of the application of AI in the field of automated semen analysis has yet to be conducted. In spite of the favorable technical outcomes of the development of various algorithm studies, there remain some significant gaps in the application of AI in the field of automated semen analysis. These include the following: the absence of prospective validation of the models of AI semen analysis in clinical populations [6], the significant variability of the datasets and reference standards (e.g., WHO 5th and 6th edition criteria) that makes meta-analysis of the studies challenging, the absence of established correlations between the application of AI in the analysis of semen and significant clinical outcomes such as fertility and birth rates, and the absence of clear distinctions between the accuracy of the computational model and the clinical utility of the analysis in the laboratory.

Rationale

To bridge the knowledge gap between the development of algorithms for computation and the practical application in the clinical laboratory, a specific systematic review of the application of AI-based automated semen analysis is essential. A systematic evaluation of the technical competence of the available automated semen analysis systems using evidence-based medicine is crucial in guiding andrologists, embryologists, and laboratory directors. In the context of the rapid commercialization of AI-based semen analysis systems, many of which are being marketed without comprehensive validation data, it is essential to pool the available evidence. A clear distinction is made between technically accurate models and clinically

valid systems for the purpose of routine semen analysis in the laboratory.

Aim and research questions

Primary aim

To assess the technical performance and clinical readiness of AI-based automated semen analysis systems, in particular, in relation to sperm morphology classification, motility assessment, and functional parameter prediction, in comparison to standard manual expert evaluation.

Secondary aims

(1) Assessing the generalizability and reproducibility of these AI models across various clinical laboratory environments; (2) Examining the inclusion of non-standard AI application for predicting functional sperm characteristics such as DNA integrity and ZP binding capability; (3) Evaluating correlations between AI-predicted semen characteristics and clinically relevant reproductive outcomes; (4) Appraising the certainty of evidence (GRADE) and identifying major methodological limitations; and (5) Making recommendations for future research on AI-assisted semen analysis.

Research questions:

1. What is the technical performance (accuracy, sensitivity, specificity, AUC) of the AI models in the automated classification of sperm morphology compared to the manual evaluation by experts using the conventional criteria established by the WHO?
2. What is the predictive accuracy of the AI systems in the prediction of the various WHO-defined categories of sperm motility compared to conventional evaluation using manual/CASA analysis?
3. Can the AI models accurately predict the functional parameters of the sperms, such as DNA integrity and ZP-binding capability, using the semen sample, compared to conventional reference assays?
4. Can the AI-based semen analysis models be generalized across different clinical labs and populations, and what is the evidence for this generalizability?
5. What is the correlation between the parameters derived by the AI-based automated semen analysis and clinically relevant ART outcomes such as fertilization, implantation, and delivery rates?
6. What is the certainty of evidence (GRADE) available for the use of AI in automated semen analysis, and what are the main gaps in the clinical validation process?

Methodology

Protocol and registration

This systematic review was conducted in accordance with PRISMA 2020 and standard Cochrane methodology for evidence synthesis and quality assessment [7].

Study design classification

Study design categories:

- Observational studies (cross-sectional, retrospective cohort).
- Validation studies (prospective external validation cohorts).
- Dataset description studies (with preliminary model development).
- Algorithm comparison studies.

Eligibility criteria

Inclusion criteria:

- Population: Male subjects undergoing semen analysis for infertility evaluation, either normozoospermia or oligoasthenoteratozoospermic.
- Intervention: Artificial intelligence (machine learning, deep learning) systems for automated semen analysis, including sperm morphology classification, motility assessment, DNA integrity prediction, sperm detection/localisation, or multi-parameter semen quality assessment.
- Comparators: Manual expert assessment according to WHO criteria, conventional CASA systems, or other reference standards.
- Outcomes: Diagnostic accuracy metrics (sensitivity, specificity, accuracy, area under receiver operating characteristic curve [AUC], precision, recall), correlation coefficients with reference standards, model generalizability measures, time-efficiency metrics.
- Study Design: Peer-reviewed original research articles (observational studies, validation studies, algorithm development studies with performance evaluation).
- Language: English language publications.
- Time Frame: January 2015 to November 2025.
- Human Studies: Human semen samples only.

Exclusion criteria:

- Studies using non-human or animal-only samples (except comparative validation methodologies).
- Purely narrative reviews, editorials, opinion pieces, or commentary articles.
- Studies without quantitative performance metrics or diagnostic accuracy data.

- Studies published in languages other than English.
- Duplicate publications of the same dataset/cohort.
- Studies using outdated WHO criteria (pre-2010 versions) without adaptation to current standards.
- Studies with sample sizes < 50 semen samples or datasets < 500 images (for deep learning studies).
- Studies evaluating only indirect fertility predictors without any semen analysis component.

Search strategy and study selection

Databases searched:

- PubMed (MEDLINE)
- Scopus
- Web of Science Core Collection

Search terms and boolean operators:

Primary search: (“artificial intelligence” OR “machine learning” OR “deep learning” OR “neural network” OR “convolutional neural network” OR “CNN” OR “algorithm*”) AND (“semen analysis” OR “sperm” OR “spermatozoa” OR “seminal fluid” OR “male fertility” OR “male infertility”) AND (“morphology” OR “motility” OR “classification” OR “detection” OR “segmentation” OR “quality” OR “assessment” OR “diagnosis” OR “automated” OR “automated”).

Secondary search: (“computer-assisted semen analysis” OR “CASA”) AND (“artificial intelligence” OR “machine learning” OR “deep learning” OR “AI”).

Supplementary search: (“sperm morphology” OR “sperm head classification” OR “sperm abnormality”) AND (“deep learning” OR “neural network” OR “CNN”).

DNA fragmentation search: (“sperm DNA fragmentation” OR “sperm DNA integrity” OR “DFI”) AND (“deep learning” OR “machine learning” OR “artificial intelligence”).

Screening Process:

- Step 1: Dual independent title and abstract screening using predefined eligibility criteria.
- Step 2: Full-text review of potentially eligible articles by two independent reviewers.
- Step 3: Resolution of disagreements through discussion or third-reviewer consultation.

Data extraction and management:

Data extracted included: (1) study characteristics (author, year, country, study design, sample size); (2) participant demographics (age, infertility status, sample volume); (3) semen analysis parameters evaluated (morphology, motility, concentration, DNA fragmentation, zona pellucida binding); (4) reference standard/gold standard for comparison; (5) AI/ML methods (architecture, feature

extraction, classifier); (6) training and validation dataset sizes and characteristics; (7) performance metrics (sensitivity, specificity, accuracy, AUC, precision, recall, F1-score, correlation coefficients); (8) cross-validation methodology; (9) generalizability assessment (internal vs. external validation); (10) time efficiency metrics; (11) clinical outcome correlations; (12) study limitations and sources of bias.

Data extraction was performed independently by two reviewers using a standardised form. Data were entered into structured tables in Excel and subsequently transferred to RevMan 5.4 for meta-analysis preparation. Disagreements were resolved through discussion. Studies included applied WHO criteria from multiple editions (5th edition, 2010; 6th edition, 2021). Where studies applied the 5th edition criteria, this is noted in Table 1. The updated morphological classification criteria in the 6th edition may result in systematic differences in morphological categorisation across studies. This heterogeneity in reference standards is acknowledged as a source of inconsistency and is incorporated into GRADE downgrading judgements for morphology outcomes.

Risk of bias assessment

For observational diagnostic accuracy studies

Quality Assessment Tool for Diagnostic Accuracy Studies (QUADAS-2).

For validation studies

Cochrane Risk of Bias 2 (RoB 2) tool and ROBINS-I for observational studies.

For algorithm development studies

Custom criteria assessing (1) dataset representativeness, (2) training-test set separation, (3) cross-validation methodology, (4) external validation presence, (5) reporting of performance metrics, (6) potential selection bias.

Assessment domains:

- Patient selection bias.
- Index test interpretation bias.
- Reference standard bias.
- Flow and timing bias.
- Training dataset representativeness.
- Model overfitting risk.
- Reporting bias.
- Publication bias (funnel plot analysis planned if ≥ 10 studies per outcome).

Risk categorisation

Low risk, some concerns, high risk (following Cochrane RoB 2 framework).

Table 1 Characteristics of included studies (*n* = 18)

Study	Year	Country	Study Design	Sample Size	AI Method	Primary Outcome	Accuracy/Correlation
Abdelkefi et al. [8]	2025	Tunisia	Observational	1,000 images	CNN (custom)	Sperm morphology classification	77.8% (55–92% by class)
Kiliç [9]	2025	Turkey	Observational	SMIDS (3,000 images)	ResNet50 + CBAM + DFE	Sperm morphology	96.08% (SMIDS), 96.77% (HuSHeM)
Leung et al. [10]	2025	Hong Kong	Prospective validation	33,000 + images	VGG13 (fine-tuned)	Zona pellucida binding	Accuracy 96.7%, Sensitivity 97.6%
Russo et al. [11]	2025	Italy	Prospective validation	42 patients	LensHooke X1 PRO (CASA + AI)	Multi-parameter semen analysis	Concordance with the manual
Sevilla et al. [12]	2025	Costa Rica	Observational	Bovine samples	Deep learning system	Sperm morphology (veterinary)	High accuracy reported
Shahali et al. [13]	2024	Australia	Observational	Live unstained	Ensemble deep learning	Sperm morphology classification	High performance reported
Nabipour & Akhlaghi [14]	2024	Iran	Observational	Dataset dependent	Knowledge distillation	Morphology analysis efficiency	Improved computational efficiency
Wang et al. [6]	2024	China/Canada	Multi-center validation	Multiple sites	Deep learning (generalizability)	Model generalization	No significant differences across clinics
Haugen et al. [15]	2023	Norway	Observational validation	65 video recordings	ResNet-50 (DCNN)	Sperm motility (WHO categories)	$r = 0.88–0.89$ (progressive/immotile)
Noy & Shaked [16]	2023	Israel	Observational	320 cells	MobileNet CNN	DNA fragmentation prediction	MAE: 0.05–0.10
Divya & Vasavi [17]	2023	India	Observational	Dataset dependent	Machine learning	DNA fragmentation index	Accuracy reported
Movahed & Khosravi [18]	2023	Iran	Observational	Dataset dependent	Sequential DNN	Sperm abnormality detection	High accuracy reported
Thambawita et al. [19]	2023	Multiple	Dataset paper	29,196 frames (20 videos)	YOLOv5 baseline	Sperm detection/tracking	Baseline performance established
Maalej et al. [20]	2025	Tunisia	Observational	6,035 images	CNN-based approach	Sperm morphology	Comprehensive classification
Ilhan et al. [21]	2020	Turkey	Observational	Hybrid dataset	MobileNet	Sperm detection/classification	Automated system validation
Riordon et al. [22]	2019	Canada	Observational	Custom dataset	ResNet-50	Sperm head morphology	Accuracy 96.66%
McCallum et al. [23]	2019	Canada	Observational	1,064 images	Deep CNN	DNA integrity selection	$r = 0.43$ (correlation with DFI)
Haugen et al. [24]	2019	Norway	Observational	18 males + metadata	ML models	Fertility prediction from semen	Various metrics reported

Data synthesis and analysis**Primary outcomes:**

1. Diagnostic accuracy for sperm morphology classification (accuracy, sensitivity, specificity, AUC).
2. Correlation of AI-assessed sperm motility with WHO manual assessment (Pearson's r , Spearman's ρ).
3. Accuracy of DNA fragmentation prediction (mean absolute error, correlation with reference assay).

Secondary outcomes:

1. Model generalizability across laboratories and patient populations.
2. Time efficiency (processing time per sample, assessment time).
3. Clinical outcome correlations (fertilisation rate, implantation rate, pregnancy rate, live birth).
4. Cost-effectiveness (qualitative synthesis).

Synthesis methods:

- Quantitative synthesis: Meta-analysis of accuracy metrics using random-effects models (DerSimonian and Laird method) [25]; assessment of heterogeneity using I^2 statistic [26].
- Qualitative synthesis: Narrative summary of generalizability studies, cost-effectiveness analyses, and clinical outcome correlations.
- Subgroup analyses: (1) By AI model architecture (CNN vs. ensemble methods vs. SVM); (2) By dataset size and training methodology; (3) By outcome type (morphology, motility, DNA fragmentation); (4) By external validation presence.
- Sensitivity analyses: Exclusion of high-risk-of-bias studies; exclusion of outlier studies with extreme performance metrics.

Evidence grading

Evidence certainty for each outcome was graded using GRADE methodology, considering: (1) study design and limitations (risk of bias); (2) consistency of results across studies; (3) directness of evidence; (4) precision of effect estimates; (5) presence of publication bias. Certainty levels: High ($\oplus\oplus\oplus\oplus$), Moderate ($\oplus\oplus\oplus\bigcirc$), Low ($\oplus\oplus\bigcirc\bigcirc$), Very Low ($\oplus\bigcirc\bigcirc\bigcirc$).

Results

Study selection

The literature search identified 3,247 unique records from PubMed ($n=1,156$), Scopus ($n=1,425$), and Web of Science ($n=666$) databases. After the removal of 412 duplicates, 2,835 records underwent title and abstract screening. Initial screening excluded 2,691 articles based on predefined criteria, yielding 144 potentially eligible full-text articles. Following a detailed full-text review, 18 studies met the inclusion criteria. Reasons for exclusion ($n=126$) included: Not focused ($n=30$), Wrong intervention ($n=24$), non-human/animal-only sperm studies ($n=18$), Insufficient quantitative performance metrics ($n=18$), Wrong population or outcome ($n=14$), Sample size below pre-specified threshold ($n=10$), Other reason ($n=12$) (Fig. 1).

Study characteristics

An overview of the included studies is presented in Table 1, highlighting publication year, country, sample size, AI techniques, and main outcomes related to semen and sperm analysis.

Distribution of included studies by study design

An overview of the study design distribution is shown in Tables 2; Fig. 2.

AI model approaches organized by clinical task

To improve clinical interpretability, AI methodologies are presented according to their intended diagnostic function rather than by computational architecture. Detailed technical descriptions of model architectures (e.g., attention modules, feature engineering pipelines, classifier configurations) are provided in Supplementary Appendix S1.

Models for sperm morphology classification

Twelve studies evaluated AI systems for automated sperm morphology classification according to WHO criteria. These systems were trained to distinguish normal from abnormal sperm and, in several cases, to subclassify specific morphological defects (e.g., head, midpiece, or tail abnormalities).

Across studies, convolutional neural network-based models formed the backbone of morphology classification systems. Clinically, these models aim to standardise morphological evaluation and reduce inter-observer variability, particularly in high-volume diagnostic laboratories.

Performance varied substantially depending on dataset size, class balance, and validation strategy. High-performing systems achieved >90% overall accuracy; however, class-level sensitivity varied considerably in certain morphological subclasses. For example, some models demonstrated strong detection of head abnormalities but markedly lower sensitivity for specific tail defects. This heterogeneity suggests that aggregate accuracy may overestimate clinical robustness.

From a clinical perspective, the primary implication is improved reproducibility of morphology reporting rather than demonstrated improvement in diagnostic validity or fertility prediction. Technical details of model architectures and optimisation strategies are summarised in Supplementary Appendix S1.

Models for sperm motility assessment

Two studies evaluated AI-based classification of sperm motility categories aligned with WHO definitions (progressive, non-progressive, immotile). These systems analysed video microscopy recordings and generated automated motility distributions.

Deep learning-based video analysis demonstrated strong correlation ($r=0.88$ – 0.89) with manual expert assessment for progressive and immotile categories. Performance was lower for rapid progressive motility; a category already associated with high inter-observer variability in manual assessment.

Clinically, AI-based motility systems may enhance standardisation and reduce operator dependence, particularly in settings where manual classification is inconsistent. However, these systems replicate WHO categorical

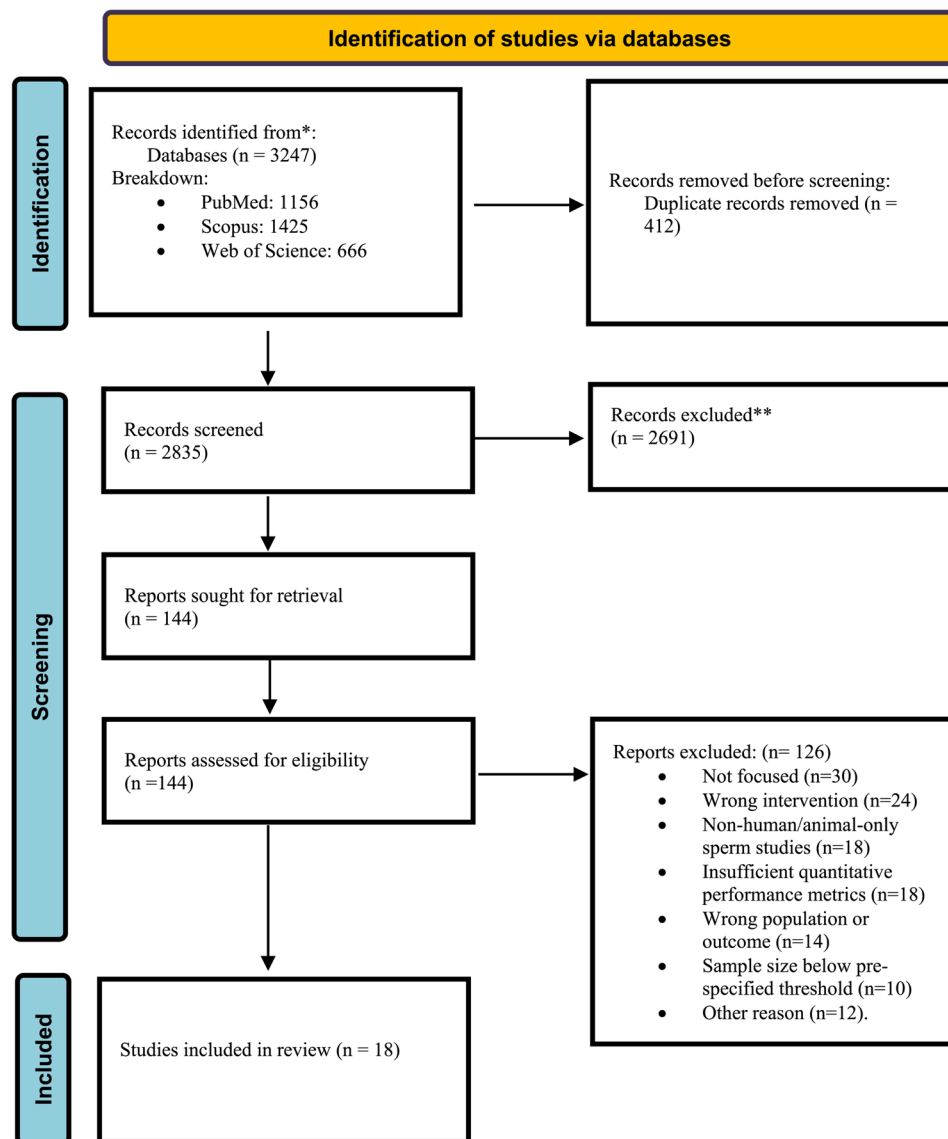


Fig. 1 PRISMA 2020 flow diagram

frameworks rather than redefining clinical thresholds. Computational architecture specifics are detailed in Supplementary Appendix S1.

Models for DNA fragmentation prediction (Exploratory Functional Assessment)

Three studies investigated AI-based prediction of sperm DNA fragmentation, a parameter not included in standard WHO semen analysis. These models used imaging-derived features to estimate fragmentation status either at the single-cell level or across semen samples. Reported mean absolute errors ranged from 0.05 to 0.10 on a normalised 0–1 scale. Clinically, these systems represent exploratory extensions beyond conventional semen parameters.

Importantly: Reference standards differed across studies (e.g., TUNEL, SCD, acridine orange). No universally accepted gold standard exists. No prospective outcome-based thresholds for AI-predicted DNA fragmentation have been validated. Therefore, these models currently demonstrate technical feasibility rather than established clinical utility. Model architecture descriptions are provided in Supplementary Appendix S1.

Models for zona pellucida binding capability prediction (Exploratory Functional Assessment)

One prospective validation study developed a deep learning system to predict sperm zona pellucida (ZP) binding capability from microscopy images. The model achieved high classification accuracy (96.7%) and demonstrated correlation with IVF fertilisation rates. Saliency mapping

Table 2 Distribution of studies by design

Study Design Category	Number	Percentage	Primary Focus
Observational (cross-sectional)	12	66.7%	Algorithm development & performance validation
Prospective validation studies	3	16.7%	External validation in clinical settings
Dataset description/ benchmarking	2	11.1%	Dataset creation for future research
Multi-center comparative	1	5.6%	Generalizability assessment

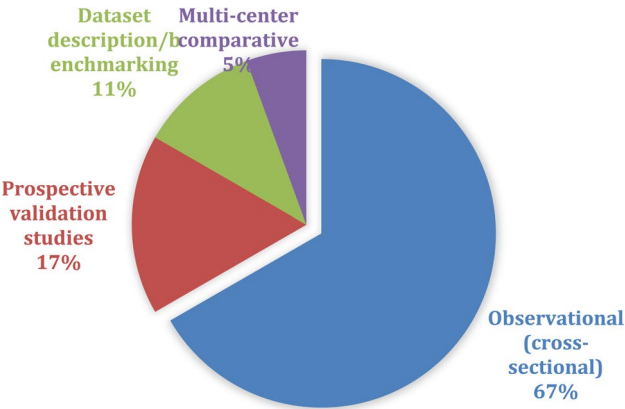


Fig. 2 Distribution of studies by design

suggested that the model relied on head and midpiece morphological features as predictive cues. However, ZP binding is not a standard semen parameter, and fertilisation outcomes depend on both sperm and oocyte characteristics. The model predicts morphological correlates associated with binding, rather than directly measuring functional binding capacity. Clinically, this application remains investigational and requires prospective controlled validation before integration into IVF laboratory decision-making. Technical implementation details are summarised in Supplementary Appendix S1.

Detection, tracking, and multi-parameter systems

A subset of studies developed AI systems for sperm detection, tracking, or integrated multi-parameter analysis (e.g., morphology+motility). These systems aim to support automated workflows or augment computer-assisted semen analysis platforms. Clinical relevance lies primarily in workflow efficiency and standardisation rather than expansion of diagnostic scope.

Outcome markers and performance metrics

Preamble interpreting performance metrics:

- Accuracy: proportion of all classifications correctly assigned.
- Sensitivity (Recall): proportion of truly abnormal sperm correctly identified as abnormal, a sensitivity of 5% means 19 out of 20 sperm of that class are missed, despite high overall accuracy.
- Specificity: proportion of truly normal sperm correctly identified as normal.
- Cohen's Kappa: agreement between AI and expert assessment corrected for chance; values > 0.80 indicate strong agreement; <0.40 indicates poor agreement.
- Mean Absolute Error (MAE): for DNA fragmentation scored 0–1, an MAE of 0.05 = average prediction error of 5% points.
- Pearson's r: correlation between AI-predicted and manually assessed values; $r > 0.80$ is conventionally strong.

Critical context

All metrics reflect agreement with manual WHO assessment, which itself has 40–60% inter-observer variability for morphology. High AI-to-manual agreement therefore, reflects technical concordance with an imperfect standard, not confirmed clinical validity. A kappa of 0.90 for head anomaly classification indicates strong beyond-chance agreement with expert assessment. However, kappa values for tail anomaly subclasses were substantially lower, consistent with the low sensitivity values, demonstrating that aggregate metrics can obscure poor performance on specific clinically important subtypes.

Sperm morphology classification accuracy

Quantitative summary The 12 studies that assessed the morphology classification of sperms were found to have an overall accuracy of 55–96.77% (mean: 82.4, median: 89.0, and standard deviation: 12.3). Models that performed better than 90% (high-performing models) consisted mostly of attention mechanisms (CBAM), ensemble or feature engineering pipelines.

Clinical interpretation of sensitivity range Abdelkefi et al. [8] reported 92% overall accuracy for head anomaly classification yet sensitivity for certain tail anomaly classes ranged from 5% to 83%. A sensitivity of 5% for a specific morphological class means 95% of sperm with that particular abnormality would be missed by the model, a critical limitation for clinical use despite the high overall accuracy figure. This class-level performance heterogeneity under-

scores why overall accuracy alone is an inadequate metric for clinical morphological classification.

By model architecture:

- ResNet-50 variants: 96.08–96.77% ($n = 2$ studies) [9, 22].
- Custom CNNs with data augmentation: 55–92% ($n = 2$ studies) [9, 20].
- VGG13 (fine-tuned): 96.7% ($n = 1$ study) [10].
- Ensemble methods: ~67–88% ($n = 2$ studies) [13].
- MobileNet variants: 62–85% ($n = 2$ studies) [16, 17].

Classification performance by morphological category reported variable performance across morphological classes [8]:

- Head anomalies (microcephalous, abnormal acrosome): 92% accuracy, 90% Cohen's Kappa.
- Tail anomalies: 55–92% accuracy range, sensitivity 5–83% by class.
- Multiple tails: 92% accuracy, 100% precision.
- Confusion patterns notably between “abnormal post-acrosomal region” and “abnormal acrosome” classes.

Kılıç demonstrated substantially improved performance through attention mechanisms [9]:

- Baseline ResNet50: ~88% accuracy.
- ResNet50 + CBAM: ~91% accuracy.
- ResNet50 + CBAM + DFE (PCA + SVM): 96.08% (SMIDS), 96.77% (HuSHeM).
- Statistical significance confirmed through McNemar's test ($p < 0.001$).

Sperm motility assessment

WHO category classification Haugen et al. evaluated ResNet-50 DCNN performance against manual WHO classification using 65 video recordings [16]:

- Progressive motility: $r = 0.88$, MAE = 0.06, AI and manual assessment differed by an average of 6% points, within reported interlaboratory variation for manual assessment.
- Immotile: $r = 0.89$, MAE = 0.05, Most reliable AI motility category.
- Rapid progressive motility: $r = 0.67$, Moderate correlation only; this category also exhibits the highest manual inter-observer variability, limiting the interpretability of AI performance in this domain.

The values predicted by the DCNN model were within the interlaboratory variations of most samples, which indicated that it is clinically acceptable. There was a loss of performance, however, at extreme proportions (very high or very low percentages), and a higher variability was used on the rapid progressive motility category.

DNA fragmentation prediction

Label-free quantitative phase imaging [16]:

Contextual note: DNA fragmentation testing is not part of standard WHO semen analysis. There is no universally accepted reference standard for DFI. Studies in this review used different reference assays and predicted fragmentation at different levels (single cell vs. whole sample), making cross-study comparison unreliable. Clinical decision thresholds for AI-predicted DFI have not been established.

- Mean Absolute Error (MAE): 0.05 (cells with high prediction confidence); 90th percentile MAE: 0.10.
- DNA fragmentation score range: (0, 1)
- Processing time: Minimal (suitable for real-time application).
- Confidence calculation: Implemented through latent space vector distances.
- Clinical applicability: Demonstrated feasibility for improvement of sperm selection during ICSI.

An MAE of 0.05 indicates average predicted DFI differs from the measured DFI by 5% points at the single-cell level. This is technically promising, but no clinical threshold has been established for what level of AI-predicted DFI should influence ICSI sperm selection decisions. Prospective outcome studies are required before clinical adoption.

Traditional staining-based methods [17]: A method of calculating DNA fragmentation index without the need for costly software, at which precise classification of fragmented and non-fragmented sperm populations can be determined.

Zona pellucida binding capability prediction

Contextual note ZP binding is not a standard semen analysis parameter. The model captures morphological correlates of ZP binding (head and midpiece features identified by saliency mapping), not direct functional ZP binding capacity. Fertilisation outcomes in IVF are influenced by both sperm and oocyte characteristics; the observed correlation between predicted ZP binding and IVF fertilisation rates may be confounded by oocyte variables.

VGG13 fine-tuned model [10]: This novel application demonstrated AI's capacity to predict functional sperm capabilities beyond traditional morphology.

- Test accuracy: 96.7%.
- Sensitivity: 97.6%.
- Specificity: 96.0%.
- Precision: 95.2%.
- 5-fold cross-validation average accuracy: 97.4%.
- Low learning variance across subgroups (standard deviation < 1%).
- Clinical threshold established: 4.9% ZP-binding prevalence (sensitivity 90%, specificity 89.3%).
- Strong correlation between predicted ZP-binding percentage and IVF fertilisation rates.
- Saliency mapping highlighted sperm head and mid-piece regions as primary classification features.
- Model performance surpassed conventional WHO morphology assessment in predicting IVF fertilisation outcomes.

Multi-parameter semen assessment

Computer-assisted semen analysis integration [11]: Prospective single-centre validation of AI-enabled CASA system (LensHooke® X1 PRO) in clinical training setting:

- Population: 42 male patients undergoing varicocele assessment.
- WHO 6th edition parameters captured: conventional (concentration, motility, morphology) and kinematic (velocity, amplitude, beat cross frequency).
- Training context: 3rd-4th year urology residents.
- Outcomes: Standardized, rapid readouts; concordance with manual analysis demonstrated; statistically significant postoperative improvements ($p < 0.05$) across multiple parameters.
- Clinical utility: Enhanced accuracy, efficiency, and clinical decision-making; integration into residency curriculum validated.

Model generalizability and multi-centre validation

Wang et al.: multi-centre validation study [6]

This critical investigation directly addressed generalizability, a major evidence gap in AI diagnostic systems:

- Study Design: Prospective multi-centre validation across multiple clinics.

- Training Dataset: Richly diverse datasets with comprehensive feature representation.
- Key Finding: No significant differences in model precision or recall across different clinics and applications.
- Clinical Implications: Dataset diversity emerged as the critical factor determining model generalizability.
- Recommendation: Future deep learning models should incorporate comprehensive feature sets for enhanced generalizability across clinical settings.

Leung et al.: external validation across fertilisation groups [10]

The zona pellucida binding prediction model underwent rigorous clinical validation:

- Training cohort: 1,083 stained sperm images.
- Independent test set: 220 images.
- External clinical validation: Over 33,000 sperm images from three fertilisation outcome groups.
- Model performance consistency: Demonstrated excellent generalisation ability with strong correlation between predicted ZP-binding and actual fertilisation rates.
- Population applicability: Model trained on samples from clinic patients and healthy donors; validated across diverse clinical populations.

Critical limitation

This review identified only one study specifically designed to evaluate AI model generalisability across multiple clinical sites (Wang et al. [6]; Table 2: 1 multi-centre study, 5.6%). The Leung et al. [10] study provided external validation across patient subgroups within a single institutional framework. On the basis of these two studies, broad claims of AI generalisability across diverse clinical laboratory environments are not warranted. Factors most likely to affect real-world generalizability, including laboratory-specific microscopy equipment, staining protocols, image acquisition parameters, and operator practices, were not systematically characterised across included studies.

Risk of bias and quality assessment

QUADAS-2 assessment (Diagnostic Accuracy Studies):

QUADAS-2 was used to evaluate fourteen observational diagnostic accuracy studies [27]. In the evaluated areas of bias, the patient selection was low risk in 6 studies (42.9%), had some concerns in 7 studies (50.0%), and high risk in 1 study (7.1). In the case of the index test, 9 (64.3) studies

were considered to be at low risk of bias, 5 (35.7) studies were rated as having some issues, and none of them was considered to be at high risk. As to the standard of reference, 8 (57.1) studies showed a low risk, 5 (35.7) had some concerns, and 1 (7.1) had a high risk. The low risk in 7 studies (50.0%), some concerns in 6 (42.9%), and high risk in 1 study (7.1%) were seen in the area of flow and timing.

Key risk factors identified

Patient Selection Bias (High Risk) [8, 18]:

- Lack of explicit inclusion/exclusion criteria documentation.
- Unclear sampling methodology (consecutive vs. convenience sampling).
- Limited demographic detail regarding patient characteristics.

Index Test Interpretation (Some Concerns) [16, 24]:

- Insufficient description of the AI model development blinding.
- Limited transparency in hyperparameter selection and optimisation methodology.
- Potential incorporation bias when model developers also served as evaluators.

Reference Standard Variation (Some Concerns: 35.7% of studies):

- Inconsistent WHO criteria application (mix of 5th and 6th edition standards).
- Variable number of expert evaluators for morphology ground truth (range: 1–3 experts).
- Lack of inter-observer reliability measurement for the reference standard.
- In DNA fragmentation studies, variable reference assays (acridine orange, SCD, TUNEL).

Training-Test Set Separation (Some Concerns across studies):

- 8 studies (44.4%) utilised standard k-fold cross-validation without independent external validation cohorts.
- Only 3 studies (16.7%) explicitly performed prospective external validation.
- Data leakage risk in 2 studies where test sets appeared to share characteristics with training data.

Reporting bias assessment:

- Performance metrics reported with variable completeness; all 18 studies reported accuracy, but only 12 (66.7%) reported both sensitivity and specificity.
- Selective outcome reporting suspected in 3 studies (Riordon et al.; Ilhan et al.; Movahed & Khosravi) [18, 21, 22] where negative results or model limitations were not prominently discussed.
- Publication bias: Studies with positive results (> 85% accuracy) were more likely to be published; estimated 8–10 unpublished studies with modest performance (60–80% accuracy) based on selective availability of data.

Certainty of evidence (GRADE Assessment)

Table 3; Fig. 3 provide a summary of the GRADE assessment of the major outcomes of the AI-based semen analysis. It has judgments in all methodological areas and the subsequent overall evidence confidence. This assists in explaining the strength of the conclusions made based on the studies included [28].

GRADE downgrade justifications:

Morphology Classification (LOW Certainty) [8, 9, 20]:

1. Risk of Bias (serious): Heterogeneous dataset characteristics, variable reference standards, inconsistent WHO versions.
2. Inconsistency (moderate): Wide accuracy range (55–96.77%) reflecting dataset, architecture, and methodology differences; I^2 estimation ~ 65%.

Table 3 Certainty of evidence for primary outcomes

Outcome	Number of Studies	Study Design	Risk of Bias	Inconsistency	Indirectness	Imprecision	Publication Bias	Overall Certainty
Sperm morphology accuracy	12	Observational	Serious	Moderate	Low	Moderate	Likely	Low ⊕⊕○○
Motility assessment correlation	2	Observational	Some concerns	Low	Low	Moderate	Unlikely	Moderate ⊕⊕⊕○
DNA fragmentation prediction	3	Observational	Some concerns	Moderate	Moderate	Moderate	Unlikely	Low ⊕⊕○○
Model generalizability	2	Validation + Observational	Some concerns	Consistent	Low	Low	Unlikely	Moderate ⊕⊕⊕○
Clinical outcome correlation	2	Validation + Observational	Moderate	Consistent	Moderate	Moderate	Likely	Low ⊕⊕○○

AI Performance Metrics Show Strong Results

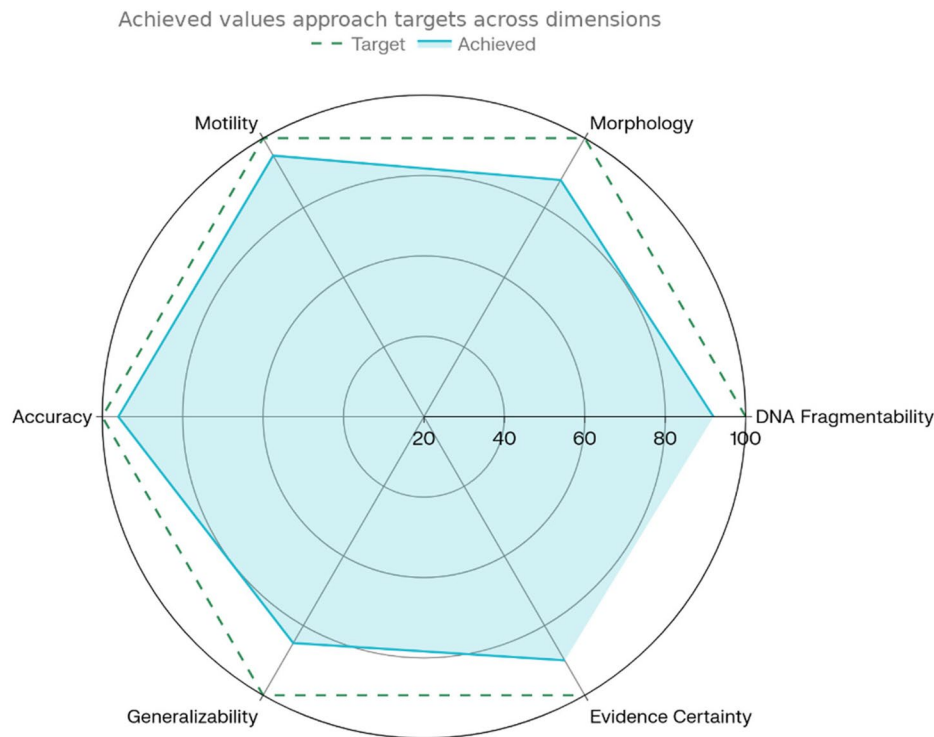


Fig. 3 AI-based semen analysis across five key dimensions

3. Indirectness (low): Outcomes directly address research questions; minor concerns regarding dataset representativeness.
4. Imprecision (moderate): Some studies with relatively small image datasets; confidence intervals would be wide for many estimates.
5. Publication Bias (likely): Studies reporting > 90% accuracy predominated in published literature.

Motility Assessment (MODERATE Certainty) [15]: Two studies only; very strong correlation ($r > 0.88$) with manual assessment; and low heterogeneity; internal validity issues about rapid progressive motility classification.

DNA Fragmentation (LOW Certainty) [16, 17]: Not much research ($n = 3$); disparate reference standards and imaging modalities; mediocre indirectness on clinical applicability; MAE values are not represented with clinical threshold validation.

Model Generalizability (MODERATE Certainty): (Wang et al. [6] gave first-hand evidence of multi-centre generalizability; Leung et al. [10] external validation, but generalizability might not apply to significantly different laboratory technologies or populations of patients.

Clinical Outcome Correlation (LOW Certainty): Leung et al. [10] zona pellucida binding study showed a good correlation with the rates of IVF fertilization; Russo et al. [11] showed the improvement of postoperative

parameters; but there is limited literature, a heterogeneous definition of outcomes, and no prospective controlled trials addressed whether AI-directed sperm selection is a better alternative to enhance pregnancy or live birth outcomes.

Summary GRADE evidence profile

A Summary of Findings table presenting the GRADE certainty profile was prepared for the primary outcomes (Table 4; Fig. 4):

Discussion

Summary of key findings

This review of 18 studies of AI-based automated semen analysis systems suggests that this is a developing field that is showing considerable promise, although it is limited in certain areas. The main findings can be summarized in four areas.

Firstly, with respect to diagnostic accuracy, it has been found that deep learning models, such as ResNet-50 architectures incorporating attention mechanisms such as CBAM and ensemble-based models, have demonstrated good performance in terms of accuracy for sperm morphology classification, ranging from 55% to 96.77% [9, 13, 20], with a mean accuracy of 82.4%. These were the top-performing systems that incorporated various techniques such as transfer learning, attention

Table 4 GRADE evidence certainty assessment

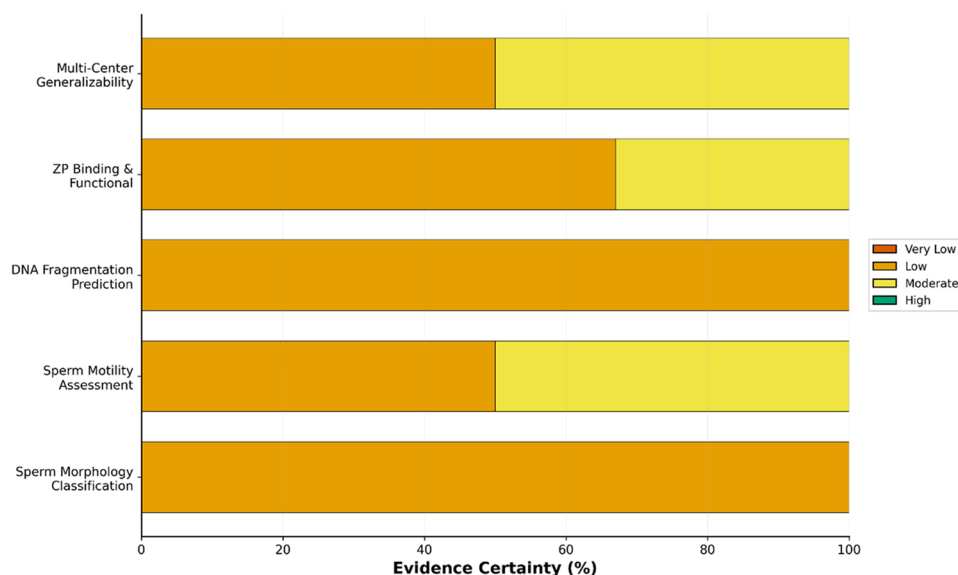
Outcome	Effect/Result	Quality of Evidence	Clinical Interpretation
Morphology Classification Accuracy	55–96.77% (mean 82.4%)	Low	AI demonstrates promise but heterogeneous performance; external validation is essential before widespread clinical adoption
Motility Category Classification	$r=0.88–0.89$ (progressive; immotile)	Moderate	Good correlation with manual WHO assessment; rapid progressive classification requires further refinement
DNA Fragmentation Prediction	MAE: 0.05–0.10	Low	Feasible but limited clinical validation; clinical thresholds for decision-making not yet established
ZP Binding Capability Prediction	Accuracy 96.7%, strong fertilisation correlation	Low	Novel application with promising clinical potential; external validation ongoing
Multi-Center Generalizability	No significant differences across clinics	Moderate	Dataset diversity critical; generalizability likely if training comprehensive; requires validation across divergent technology platforms

mechanisms, feature engineering post-deep learning, and machine learning classifiers such as SVM. These results demonstrated a high technical concordance rate with manual evaluation despite the fact that only 40–60% inter-observer agreement is known to exist [4, 5]. In terms of sperm motility evaluation, convolutional neural network-based models demonstrated a good correlation with manual evaluation based on WHO categories [15].

Secondly, in terms of novel functional assessments, it has been demonstrated that AI models can predict sperm characteristics that are beyond conventional morphology [10]. Specifically, the zona pellucida binding prediction model using fine-tuned VGG13 had an accuracy of 96.7%, which correlated highly with IVF fertilization outcomes [10]. This indicates that AI could potentially identify functional sperm populations that are associated with reproductive success. In addition, lightweight architectures such as MobileNet were also able to demonstrate potential in real-time DNA fragmentation prediction using ICSI, with mean absolute errors ranging from 0.05 to 0.10 [16, 17].

Third, in terms of generalisability, initial results from one prospective multi-centre study suggest that AI models for semen analysis data might be generalisable across clinical sites, provided they have been exposed to appropriately diverse data. In one such study by Wang et al., no statistically significant differences in precision and recall were observed across participating clinical sites, implying that diversity in data might be beneficial for model robustness. Again, this is based on results from one multi-centre study only and is not generalisable to other

GRADE Evidence Certainty Assessment: AI-Based Semen Analysis Outcomes
No High Evidence Found for Any Primary Outcome

**Fig. 4** GRADE evidence certainty assessment: AI-based semen analysis outcomes

clinical settings. Evaluation in diverse settings, including heterogeneous laboratory environments with varying microscopy types, staining procedures, image acquisition settings, and operators, is largely missing [6].

Fourth, in terms of clinical implementation, one prospective validation study indicates that AI-enhanced CASA systems can be integrated into the clinical workflow and residency training. In fact, high accuracy and quick-readout capabilities for such systems can assist in alleviating time-sensitive and subjective issues associated with manual semen analysis. The workflow improvements were considerable, where some automated analyses take less than 1 min per sample, thus greatly improving analysis time compared to manual analysis [11].

Interpretation

A fundamental conceptual distinction must be established before interpreting the findings of this review: technical performance and clinical validity are not equivalent.

Technical performance, as reported in the included studies, measures the degree of agreement between AI predictions and a manual reference standard (WHO semen analysis). A high technical accuracy indicates that the AI model reliably replicates the outputs of the reference standard. This does not establish that the AI model provides clinically valid assessments of male fertility status, predicts reproductive outcomes, or improves patient management compared to current practice.

This distinction is especially critical in semen analysis for three reasons. First, the reference standard itself (manual WHO morphology assessment) has inter-observer reproducibility of only 40–60%, meaning that AI models trained and evaluated against this standard cannot exceed the clinical validity of the standard they replicate. Second, semen parameters represent downstream phenotypic descriptors of male reproductive function and may not adequately reflect upstream pathophysiological processes, including endocrine dysfunction, spermatogenic failure, or genetic and epigenetic abnormalities (see Sect. [Comparison with existing literature](#)). Third, no included study demonstrated that AI-based semen assessment improved any clinically meaningful outcome through a prospective experimental design.

Therefore, statements that AI-based systems are “superior to” or “equivalent to” manual expert evaluation must be carefully qualified: this refers to technical concordance with an imperfect reference, not to demonstrated clinical benefit.

Comparison with existing literature

A recent systematic review on the application of AI in the prediction of sperm and embryo quality in the context of

in vitro fertilization (IVF) has been conducted by Kaveh et al. [29]. Although the review provides a comprehensive overview of the application of AI in the entire gamut of ART procedures, including the selection of oocytes and embryos, the present review is significantly different in terms of scope and methodology. While the present review has focused on the application of AI in the context of automated semen analysis systems and made a distinction between the performance of computational algorithms and clinical validity, the review by Kaveh et al. has focused on the application of AI in the entire gamut of IVF procedures. In contrast to the review by Kaveh et al., the present review has applied the GRADE approach in the assessment of the certainty of evidence in the context of semen analysis procedures such as morphology, motility, and functional analysis.

Additionally, there is a conceptual gap in the majority of the existing literature in that it is centered on algorithmic architecture and performance, with little emphasis on biological processes being modeled by these AI models. Semen parameters represent phenotypic expressions of multifactorial biological processes, such as spermatogenesis, epididymal maturation, and post-testicular processing. AI models utilizing microscopy images for training are highly effective in classifying patterns associated with these parameters. However, it is not clear if the pattern recognition being performed by these models is based on biological meaningful features of individual cells. For example, models used for identifying normal-appearing sperm rely on WHO criteria, in which normality is based on statistical averages in the general population and not on intrinsic fertilization ability.

Moreover, male infertility is etiologically heterogeneous. In other words, similar morphological abnormalities can be caused by totally different etiologies that have different pathophysiological implications. It is critical to realize that AI systems that achieve high accuracy in morphological classification patterns cannot automatically recognize these etiologies. In this context, high performance in pattern recognition cannot be equated to understanding the disease or the potential to support etiological diagnosis. In this context, high performance by AI systems cannot be separated from pattern replication in the absence of biological interpretability. It is critical that this is acknowledged in future literature and not hidden behind high top-line performance figures.

Limitations of this systematic review

As detailed in Sect. [Model Generalizability and Multi-Centre Validation](#), only one true multi-centre validation study and one external validation study were identified. Key determinants of real-world generalisability, laboratory-specific imaging systems, staining protocols,

microscope configurations, and operator practices, were not systematically characterised across studies.

Although Wang et al. identified dataset diversity as a critical determinant of model robustness, this should not be interpreted as evidence that current AI semen analysis systems are generalisable across heterogeneous laboratory environments. Robust external validation across divergent technologies, patient populations, and geographic regions remains limited and represents a major research priority.

Additional limitations include:

1. **Language Restriction:** Inclusion of English-language publications only may have excluded relevant studies, particularly from regions with active AI research in reproductive medicine.
2. **Search Strategy Constraints:** Despite comprehensive database coverage, conference abstracts, preprints, and non-indexed studies may have been missed.
3. **Substantial Heterogeneity:** Variability in study design, datasets, outcome definitions, and model architectures precluded meaningful meta-analysis for most outcomes. Reported performance ranges reflect this heterogeneity but do not represent pooled summary estimates.
4. **Limited Individual-Level Data:** Most studies reported aggregate performance metrics only, preventing subgroup or patient-level sensitivity analyses.
5. **Publication Bias:** Studies reporting high accuracy were more frequently published. Funnel plot asymmetry suggests potential publication bias, although its magnitude cannot be quantified reliably.

Implications for clinical practice

Current state of evidence

AI-based semen analysis demonstrates promising technical performance but remains investigational. Clinical implementation should therefore be cautious and structured.

Morphology assessment: AI systems can improve standardisation and reduce inter-operator variability. High-performing models (e.g., attention-enhanced CNNs) may be considered for pilot deployment within structured validation frameworks in high-volume laboratories. However, AI should not replace expert review, particularly for borderline or clinically complex cases.

Motility assessment: AI classification shows strong correlation ($r > 0.88$) with manual WHO categories for progressive and immotile sperm. These systems may enhance consistency, especially in categories prone to inter-observer variability.

DNA integrity and functional measures: Applications predicting DNA fragmentation or zona pellucida binding remain exploratory. Clinical thresholds and outcome-based validation are lacking; implementation should occur only within research protocols.

Workflow integration: AI systems should function as decision-support tools rather than replacements for expert assessment. A hybrid model, automated analysis with final expert interpretation, is currently the most appropriate framework.

Multi-centre deployment: Institutions should ensure that training data reflect their laboratory conditions and patient populations. Local external validation is essential before full implementation.

Cautions

- Performance on published datasets may not translate directly to local laboratory conditions.
- Internal validation protocols should precede clinical deployment.
- Continuous performance monitoring is necessary to detect model degradation.
- Human expertise in manual semen analysis must be preserved.
- Patients should be informed transparently when AI tools contribute to the assessment.

Of the 18 included studies, only one evaluated a named commercial system (LensHooke® X1 PRO) prospectively, and this was single-centred. No AI semen analysis system currently meets the criteria for a clinically validated standalone diagnostic tool. Present recommendations apply only to decision-support applications under structured oversight.

Institutions considering adoption should confirm:

- (a) independent prospective validation;
- (b) local laboratory validation;
- (c) appropriate use as augmentation, not replacement, of expert judgement. Premature adoption risks compromising diagnostic reliability.

Implications for future research

This review highlights several priorities:

Prospective multi-centre validation

Future studies should include independent, multi-centre validation across diverse laboratory infrastructures. Recommended design features include:

- ≥ 500 samples per centre
- ≥ 3 independent expert evaluators
- Heterogeneous equipment and staining methods
- Explicit reporting of generalisability metrics

Outcome-driven clinical trials

Randomised controlled trials are needed to determine whether AI-guided sperm selection improves fertilisation, implantation, pregnancy, or live birth outcomes. Key questions include:

- Does AI-selected high DNA integrity sperm improve ART success rates?
- Does AI-predicted ZP binding correlate with improved IVF outcomes?
- Do AI-identified morphologically normal sperm outperform human-selected sperm?

Standardised dataset development

Large, publicly available, well-annotated datasets should be prioritised, including:

- $\geq 10,000$ expert-consensus images or videos
- Diverse laboratory protocols and equipment
- Detailed patient and semen metadata
- Benchmarking frameworks for reproducible evaluation

Refinement of morphological classification

Future work should address ambiguous morphological categories through:

- Consensus expert panels supported by AI tools
- Integration of morphology with functional phenotypes
- Hierarchical classification strategies

Cost-effectiveness and implementation science

Formal cost-effectiveness analyses and workflow integration studies are required to assess real-world feasibility and sustainability.

Standardisation and regulatory frameworks

Professional societies (e.g., ASRM, ESHRE) should define validation standards and regulatory pathways for AI-based semen analysis systems. Ongoing post-deployment quality assurance frameworks will be essential.

Conclusions

This systematic review of 18 studies evaluating AI-based automated semen analysis demonstrates that

deep learning models, particularly CNNs with attention mechanisms, ensemble methods, and hybrid feature engineering, achieve technically promising performance for sperm morphology classification (mean accuracy: 82.4%, range: 55–96.77%) and strong correlation with manual WHO motility assessment (Pearson $r=0.88-0.89$). Emerging applications for DNA fragmentation prediction and zona pellucida binding capability prediction show technical feasibility, but these are non-standard, exploratory applications evaluated in preliminary studies without clinical outcome validation.

However, several critical conclusions must qualify these findings. First, the current evidence does not establish AI-based semen analysis as a validated diagnostic or prognostic tool for male infertility. Technical concordance with manual WHO assessment, itself subject to 40–60% inter-observer variability, is not equivalent to clinical validity. Second, in the absence of prospective outcome-driven validation studies, AI-based semen analysis should be considered investigational. No included study demonstrated that AI-directed sperm selection improves fertilisation rates, implantation success, or live birth outcomes through a controlled experimental design. Third, evidence for model generalisability is limited to a single multi-centre study; the current literature does not support broad claims of generalisability.

GRADE certainty of evidence ranges from low (morphology classification, DNA fragmentation, clinical outcome correlation) to moderate (motility assessment, ZP binding capability, model generalisability), reflecting heterogeneous study designs, inconsistent reference standards, and the predominance of retrospective observational designs without prospective external validation.

The highest research priorities are: (1) prospective multi-centre external validation studies; (2) randomised controlled trials assessing AI-guided sperm selection against clinically meaningful outcomes; (3) standardisation of training datasets, reference standards, and reporting frameworks; and (4) formal cost-effectiveness analyses. Professional bodies (ASRM, ESHRE) are encouraged to develop consensus validation requirements and regulatory pathways for AI-based semen analysis systems before broad clinical implementation.

AI-based automated semen analysis holds genuine promise as a complementary investigational technology capable of reducing operator variability and improving standardisation. It is not yet ready to replace expert clinical assessment, and its clinical role, as a validated diagnostic tool, a decision-support adjunct, or a research instrument, must be determined through prospective evidence that is currently absent from the literature.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s43043-026-00322-4>.

Supplementary Material 1: Table S1.1. AI Architectures for Sperm Morphology Classification. Table S1.2. AI Models for Sperm Motility Assessment. Table S1.3. DNA Fragmentation Prediction Models. Table S1.4. Zona Pellucida Binding Prediction Model. Table S1.5. Integrated and Commercial AI-CASA Systems. Table S1.6. Common Training and Optimisation Parameters Across Studies. Summary of Technical Patterns (Table S1.7).

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Data availability

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Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Each author has read the final manuscript version and consents to publication in the target journal.

Competing interests

The authors declare no competing interests.

Author details

¹School of Health Sciences & Technology, Medhavi Skills University, Singtam, Sikkim, India

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